Bimodal joint population characterisation at Najd-affected road cuts, Makkah region, western Saudi Arabia

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ABSTRACT

The Najd Fault System imposes a persistent NW-SE structural corridor across the western Arabian Shield and influences rock-mass behaviour along major transport corridors. Highway rock slopes in this region are commonly described as controlled by Najd-related joints and younger NE-SW structures associated with Red Sea rifting, but these descriptions remain largely qualitative. This paper provides a structured quantitative characterisation of joint orientations at five Najd-affected road-cut sites near Makkah, western Saudi Arabia. Joint and slope dip-direction measurements in granodiorite were analysed using circular statistics, including mean direction, resultant vector length, circular variance and the Rayleigh test for non-uniformity. Each measurement was then classified into Najd, Red Sea or Other structural families using strict azimuth windows centred on the regional NW-SE and NE-SW trends. The results show that four of the five sites display statistically significant directional clustering, confirming that the observed orientations are structurally organised rather than random. All sites contain coexisting Najd-parallel and Red Sea-parallel joint families, with site-specific proportions and a substantial residual Other population. The workflow converts the qualitative statement that Najd and Red Sea fabrics control road cuts into measurable, repeatable indicators that can support later development of a Najd Intensity Index, rock-mass quality interpretation, slope-risk screening and probabilistic stability assessment.

1. Introduction

1.1 Practical problem

Major highway and infrastructure projects in western Saudi Arabia cut through crystalline basement that is heavily structured by the Najd Fault System and, more locally, by Red Sea–related faults. Designers and operators know, from experience, that some cuts are systematically more problematic than others: blocks detach along persistent joints, wedge failures develop where two sets intersect, and certain alignments repeatedly require heavier support.

In most reports, these issues are described in geological shorthand: slopes are said to be "controlled by Najd joints" or "influenced by Najd and Red Sea fabrics". While these descriptions are accurate at a conceptual level, they remain qualitative. They do not provide a measurable, site-level split between Najd-aligned, Red Sea-aligned, and other structural families that can be used directly in indices, screening tools, or probabilistic models.

1.2 Missing structure

The missing piece is a simple, defensible framework that takes real field measurements—joint and slope dip directions at actual road cuts—and answers three basic questions:

- Are orientations at these sites statistically clustered, or close to random?
- Do they form a consistent bimodal pattern that can reasonably be linked to Najd and Red Sea trends?
- How much of each site's joint population belongs to Najd, to Red Sea, and to "Other" orientations, in percentage terms?

Without this structure, later steps in the Najd research programme (Najd Intensity Index, Deltatheta zonation, probabilistic slope stability) would rest on informal impressions rather than measurable inputs. The methods adopted here are designed to be replicable from a finite number of accessible field surfaces, producing outputs that bridge raw observations and structured, statistically grounded indicators for later decision-making.

1.3 Objectives

The objectives of Paper 1 are to:

1. Test whether joint orientations at five Najd-affected road-cut sites near Makkah are significantly non-uniform using circular statistics.
2. Determine whether these orientations form a bimodal pattern consistent with Najd (NW–SE) and Red Sea (NE–SW) families at each site.
3. Quantify the proportion of measurements assigned to Najd, Red Sea, and Other classes at each site, as an explicit input to later indices and risk tools.

The emphasis is practical: the outputs are designed to be reused in later phases of the Najd programme and, eventually, in project-level decision frameworks for road alignment and support design.

2. Geological and engineering setting

2.1 Regional context

The Najd Fault System forms a NW–SE striking belt more than 1,100–1,200 km long and up to 300 km wide across the northern Arabian Shield. It comprises a braided network of left-lateral strike-slip faults, ductile shear zones, and brittle fracture corridors that overprinted earlier accretion structures during late Pan-African evolution.

In the western Shield, Najd-related fabrics cut granitic and metavolcanic rocks that now host highways, tunnels, urban expansion, and strategic underground works. Superposed on this older fabric is a younger NE–SW structural grain associated with Red Sea rifting, margin uplift, and related faulting. Together, these trends define the main tectonic directions we expect to see in slope-scale joints and small faults.

2.2 Study area and sites

The five study sites are road cuts in granodioritic basement within the Makkah region of western Saudi Arabia. Each site consists of one or more steep slopes exposing jointed and, locally, sheared crystalline rock. The key common features are:

- Lithology: granodiorite and related plutonic rocks, providing a relatively uniform intact-rock background.
- Slope geometry: high, steep cuts where joint persistence and orientation strongly influence block size and failure kinematics.
- Structural setting: within the mapped Najd belt and within tens of kilometres of NE–SW Red Sea–related lineaments.

Keeping lithology roughly constant allows this paper to focus on structural orientation as the main variable, leaving rock-mass quality indices (GSI/RMR) for later work.

2.3 Engineering relevance

From an engineering-geology point of view, the Najd system is a regional predictor of rock-mass performance: it fragments massive crystalline rocks into blocks, organises joint sets into persistent families, controls groundwater pathways, and constrains the likely kinematic modes of failure. Red Sea–related structures add another family of discontinuities that can support, intersect, or undercut Najd-parallel sets.

For designers, understanding whether a given road-cut site is Najd-dominated, Red Sea-dominated, or strongly mixed is not an academic curiosity; it directly affects expectations for block size, anisotropy, groundwater behaviour, and the need for structural support. This paper is the first step in turning that understanding into measurable proportions and simple statistics.

3. Data and methods

3.1 Field data collection

We collected joint and slope orientation data at five Najd-affected road-cut sites in the Makkah region. At each site, 100 measurements of dip direction and dip were taken using a standard compass–clinometer, following engineering-geology scanline practice, for a total of 500 measurements.

Scanlines were positioned to sample the full exposed height and lateral extent of each cut, aiming to avoid visual bias toward any single dominant set. Where access allowed, measurements were taken on benches as well as from the road shoulder to capture mid-slope surfaces. Wet or heavily weathered surfaces were cleaned locally so that readings reflected the underlying joint plane rather than surface roughness or weathering crusts.

3.2 Orientation conventions and data checks

Orientations are treated as dip-direction azimuths on the range 0–360°, measured clockwise from geographic north. Strike values were used only for internal consistency checks and are not analysed separately.

Before analysis, the dataset was screened for obvious recording errors such as swapped dip and dip-direction columns or dip angles exceeding 90°. Suspect entries were cross-checked against field notes and photographs; measurements that could not be corrected with confidence were removed. This represents only a small fraction of the data and does not change the overall orientation patterns.

3.3 Circular statistics

Dip directions lie on a circle, so linear statistics are not appropriate. For each site, we therefore computed:

- The mean dip direction;
- The resultant vector length R, a measure of concentration;
- The circular variance, derived from R.

We then applied the Rayleigh test at each site to test the null hypothesis of a uniform distribution on the circle. A small p-value ($p < 0.05$) is interpreted as evidence that orientations at that site are significantly clustered, consistent with structural control by one or more preferred joint sets.

Site-level results are reported as N, mean direction, R, and Rayleigh p-value. These metrics distinguish clearly clustered sites from sites where orientations are closer to random, and they provide a simple, reproducible way to describe “how strong” the structural signal is at each location.

3.4 Bimodal classification into Najd, Red Sea, and Other

To link these statistics to regional tectonics, each measurement was classified into one of three structural families using strict azimuth windows.

- Najd-related directions were defined by two opposite windows centred on the NW–SE Najd trend: 310–340° and 130–160°.

Red Sea–related directions were defined by two opposite windows centred on the NE–SW trend: 40–70° and 220–250°.

Each dip-direction was assigned as:

- Najd if within either Najd window;
- Red Sea if within either Red Sea window;
- Other otherwise.

The window limits are deliberately narrow. They are chosen to capture clearly aligned measurements and to avoid forcing ambiguous directions into either tectonic family. As a result, the Najd and Red Sea percentages reported here are conservative: if a measurement does not strongly align with either trend, it is counted as “Other” rather than inflating one family.

For each site we computed the percentage of measurements in each class, producing a three-way composition (Najd, Red Sea, Other) that can be compared across sites and used as an input to later indices and decision tools.

3.5 Sensitivity and uncertainty

To test the robustness of the classification, we carried out a simple sensitivity analysis. All window boundaries were shifted by ±5–10°, and the Najd, Red Sea, and Other percentages were recomputed. The qualitative pattern—persistent Najd and Red Sea families at all sites and a substantial residual Other class—remained stable across these shifts.

With N = 100 per site, binomial confidence intervals can be attached to each class proportion. In this paper, we report point estimates and interpret them as site-level indicators rather than exact population values. Later probabilistic work can formalise these intervals where necessary for design.

3.6 Visualisation

We used three main visual tools:

- Rose diagrams of dip directions for each site, showing the concentration around Najd and Red Sea trends.
- Mean-vector compass plots summarising site-level mean directions on a single diagram.
- Stacked bar charts of Najd, Red Sea, and Other proportions for each site, backed by a summary table listing N, mean direction, R, Rayleigh p, and class percentages.

These visualisations are designed to be reused in presentations and in later papers as part of a consistent “Najd toolkit”.

4. Results

4.1 Non-uniformity and clustering

The Rayleigh test shows that four of the five sites have statistically significant clustering of dip directions ($p < 0.05$), with Site 3 displaying the strongest concentration ($R \approx 0.735$). Site 5 has a much lower resultant length ($R \approx 0.14$) and, at first glance, appears close to uniform.

This pattern confirms that in most sites the joint orientations are not random but organised around one or more preferred directions.

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Even at Site 5, the apparent randomness at the pooled level breaks down into more interpretable structures once the bimodal classification is applied.

4.2 Bimodal compositions by site

Using the strict azimuth windows, all five sites show coexisting Najd and Red Sea families plus a residual Other class. Based on the current dashboard values, example compositions are:

Table 1 Bimodal structural-family composition by site based on strict azimuth-window classification

Site	Najd (%)	Red Sea (%)	Other (%)
Site 1	28	30	42
Site 2	50	18	32
Site 3	38	12	50
Site 4	26	27	47
Site 5	32	8	60

Across sites, Najd-aligned directions are always present and typically form the largest or joint-largest single class, consistent with the regional dominance of the Najd structural corridor. Red Sea-aligned directions are systematically present but generally less abundant. The Other class remains substantial, with 30–60% of measurements at most sites falling outside the strict Najd and Red Sea windows.

4.3 Site-specific patterns and mixing

Site 2 stands out as Najd-dominated, with about half of all measurements falling in the Najd windows and a smaller but clear Red Sea component. Site 1 and Site 4 show more balanced Najd–Red Sea proportions, indicating stronger mixing or comparable influence of both families.

Site 3 combines strong clustering with a significant Other population, suggesting that additional structural sets or local geomorphic controls contribute to the orientation pattern. Site 5, with the highest Other percentage and low resultant length, appears to be a three-way mixture where Najd and Red Sea trends are present but not dominant.

5. Discussion

5.1 Structural interpretation

The coexistence of NW–SE and NE–SW joint families at all five sites matches the regional picture of a Proterozoic Najd shear corridor later overprinted and reactivated by Red Sea rifting. Within this framework:

- Najd-parallel joints represent the preserved and locally reactivated fabric of the older shear system.

Red Sea–parallel joints represent younger NE–SW structures, including reactivation of favourably oriented Najd planes and new faults.

The substantial Other category is not a residual "error" but a structural space that likely includes: other Precambrian shear systems, local faults not aligned with the main corridors, bedding or foliation where present, and slopes where valley geometry and road alignment exert strong control. Recognising this class explicitly keeps the analysis honest: the paper does not force every measurement into "Najd vs Red Sea" when the rock mass is more complex.

5.2 Qualitative-to-quantitative bridge

At a conceptual level, engineers and geologists in the region already acknowledge that "Najd and Red Sea control these slopes". The contribution of this paper is to turn that statement into numbers:

- A testable measure of clustering at each site (R, Rayleigh p).
- A three-way split of joint orientations into Najd, Red Sea, and Other, site by site.

This bridge is important because later steps—Najd Intensity Index, Deltatheta zonation, probabilistic FS—need numerical inputs, not only structural narratives. By keeping the windows strict and the Other category explicit, the paper also preserves transparency about what is known and what remains unresolved.

5.3 Implications for engineering decisions

From a decision perspective, the main outputs of this paper can be used in three ways:

1. Site-level structural profiles: Designers can classify a given cut as Najd-dominant, Red Sea-influenced, or strongly mixed, based on measured proportions rather than impressions.
2. Input to Najd Intensity Index: The Najd and Red Sea percentages provide a direct structural input to the composite Najd Intensity Index developed in Paper 2, which links structural intensity to GSI/RMR and Hoek–Brown parameters.
3. Screening logic for alignments: In combination with slope aspect and dip, the presence of strong Najd or Red Sea families can be used to flag more hazardous alignment–slope combinations before detailed design, forming the basis of later traffic-light maps.

In other words, this paper does not stop at confirming a bimodal pattern; it frames that pattern explicitly as a measurable, reusable input for subsequent decision tools.

6. Conclusions

1. Joint orientations at four out of five Najd-affected road-cut sites in the Makkah region are significantly clustered according to the Rayleigh test, confirming non-uniform structural control.
2. All five sites host coexisting Najd (NW–SE) and Red Sea (NE–SW) joint families, alongside a substantial residual Other class that captures additional structural and geomorphic influences.
3. Site-level compositions show Najd generally dominant, with locally stronger Red Sea influence at some sites and strong three-way mixing at others; this variability is itself a useful indicator of local structural history and risk context.
4. By converting qualitative statements about Najd and Red Sea control into simple, repeatable metrics, this paper provides the structural foundation for the Najd Intensity Index and for later decision tools that link structural fabric to rock-mass quality, slope risk, and probabilistic factors of safety.

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